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ADAPTIVE NETWORK TRAFFIC ANOMALY DETECTION USING LIGHTWEIGHT ONLINE STATISTICAL METHODS

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ABSTRACT

Network intrusion detection must flag abnormal traffic — volumetric floods, port scans, and shifts in the protocol mix — quickly and without overwhelming an operator with false alarms. Single-feature threshold monitors are cheap but structurally blind to any anomaly that does not move traffic volume, while learning-based detectors are accurate at the cost of labelled training data, memory, and periodic retraining. This paper asks whether a lightweight, training-free statistical detector can occupy the middle ground: catch the anomalies that volume monitors miss, keep the false-alarm rate low, and remain accurate when normal traffic slowly changes. We develop a reproducible window-level traffic simulator with five labelled anomaly families and a matching pipeline over the real CIC-IDS2017 flow dataset. Against five baselines — fixed threshold, moving average, z-score, a static Mahalanobis-distance detector, and an Isolation Forest reference — we evaluate a proposed online detector that maintains an exponentially weighted mean and covariance of a small multi-feature vector, with a gated update that prevents a suspected attack from being absorbed into the model of normal behaviour. On thirty held-out simulated traces the proposed detector matches the strongest static baseline in stationary conditions (event F_1 of 0.924 versus 0.917, $p = 0.90$) and significantly outperforms it when benign traffic drifts (0.914 versus 0.796 at the strongest drift, a gain of 0.118, Wilcoxon $p = 0.002$), while beating every single-feature baseline by 0.19 to 0.37 F_1 ($p < 10^{-4}$). On real CIC-IDS2017 traffic the multivariate detectors identify the non-volumetric port scan almost perfectly (ROC-AUC 0.999, recall 1.0) where volume-only monitors miss most of it (recall 0.4). An ablation shows that the gated update and the full covariance carry the performance. The detector needs no training and keeps only 72 floating-point values of state, roughly 167 times less than the Isolation Forest.

Keywords: network anomaly detection, traffic monitoring, online learning, Mahalanobis distance, EWMA, non-stationarity, CIC-IDS2017, lightweight statistical methods.

1. INTRODUCTION

Modern organisations depend on networks for nearly every service they run, which makes the network an attractive target. A network intrusion detection system (NIDS) observes traffic and raises an alert when it sees behaviour that looks malicious. *Signature-based* systems match traffic against a database of known attack patterns; they are precise on known attacks but cannot recognise an attack for which no signature exists. *Anomaly-based* systems instead

build a model of normal behaviour and flag whatever deviates from it, which in principle allows them to catch previously unseen attacks.

The practical difficulty with anomaly-based detection is rarely the dramatic event. A tenfold surge in packets is caught by the crudest threshold. The hard cases are the quiet ones. A port scan contacts many destination ports while sending very little traffic, so a monitor that watches only volume never sees it. A shift in the protocol mix — an unexpected rise in the share of UDP, say — moves no volume at all. At the same time, ordinary but bursty activity such as a flash crowd can trip a naive detector into a stream of false alarms. A useful detector therefore has to be sensitive to several facets of traffic at once, and calm in the face of benign variation. A second difficulty is that normal traffic is not stationary: over hours and days the baseline drifts with daily cycles and slow growth, so a detector whose notion of "normal" is fixed at calibration slowly becomes mis-calibrated.

This paper studies whether a single lightweight statistical detector can be at once training-free, sensitive to multiple features, and robust to this drift. Our contributions are: (i) a reproducible window-level traffic simulator with five labelled anomaly families and benign flash crowds that stress false-alarm behaviour; (ii) a proposed online detector that tracks a multi-feature mean and covariance with a gated update, so a suspected attack cannot pull the baseline toward itself; (iii) a careful comparison on thirty held-out traces with paired significance tests, an ablation of every design choice, and per-window cost measurements; and (iv) a matching evaluation on real CIC-IDS2017 port-scan and DDoS traffic. Our central and deliberately narrow finding is that multi-feature scoring — not online adaptation — defeats non-volumetric attacks, and that online adaptation earns its keep specifically when traffic is non-stationary.

2. RELATED WORK / LITERATURE REVIEW

Lightweight statistical monitoring of network traffic has a long history. The exponentially weighted moving average (EWMA) [4] and the cumulative sum (CUSUM) [5] control charts, drawn from statistical process control, are standard tools for detecting a shift in a monitored time series and are widely applied to volume-based network monitoring. Their appeal is their low cost; their limitation is that, applied to a single aggregate such as packets per second, they inherit the blindness of any volume-only monitor to non-volumetric anomalies.

A large body of work applies machine learning to network anomaly detection, from one-class support vector machines and autoencoders to tree ensembles and deep models; general and network-specific surveys catalogue these approaches and their trade-offs [1, 2]. These methods can represent complex, multi-feature notions of normal, but typically require a training phase and non-trivial memory, and many are evaluated in a supervised setting that quietly assumes a labelled attack set. The Isolation Forest [6], which isolates outliers with random partitioning, is a representative unsupervised member of this family and is used here as a heavier reference.

That normal traffic drifts is well recognised, and online or streaming variants of detectors address it by updating their model over time. The tension such methods must resolve is that an update rule which tracks benign change will also, unless it is careful, absorb a sustained attack into its notion of normal and go blind; the gated update at the core of our detector is a direct response. For multivariate scoring, the Mahalanobis distance provides a natural outlier measure that accounts for the spread of and correlations between features.

Finally, several studies compare anomaly detectors, but a recurring shortcoming is that they

do so on a single dataset or at a single operating point, which makes it hard to separate a property of the method from a property of the data. A related issue is benchmark quality: analyses of CIC-IDS2017 [7] have documented labelling and processing errors that can distort reported results [8], and surveys of intrusion-detection datasets discuss the criteria a sound benchmark should meet [3]. We respond to both concerns by pairing a controlled simulator with real-data validation and by including controls that let each gain be attributed to a specific cause.

3. RESEARCH GAP AND PROBLEM STATEMENT

The literature leaves a specific gap. Single-feature statistical monitors are cheap but cannot represent attacks that are not volumetric. Multivariate and learning-based detectors can, but the common multivariate baseline — a Mahalanobis model with statistics frozen at calibration — silently assumes that "normal" does not move, which real traffic violates over hours and days. Heavier detectors handle drift by retraining, which reintroduces the cost a lightweight monitor is meant to avoid. What is under-examined is whether a detector can be *simultaneously* training-free, multi-feature, and robust to slow non-stationarity — and, just as important, a clear accounting of which of those properties actually buys detection power.

Problem statement. Traffic is summarised as a sequence of fixed-length time windows, each described by a feature vector $\mathbf{x}_t \in \mathbb{R}^p$. A prefix of windows is known to be benign (a calibration period); afterwards each window is either benign or part of an anomaly event. A detector must emit a per-window anomaly score causally, using only windows up to time t, from which an alarm is produced by a threshold. The goal is to maximise the fraction of anomaly *events* detected and minimise alarms on benign windows — especially during benign flash crowds — at low and constant per-window cost, and without a supervised training phase.

4. PROPOSED METHODOLOGY

4.1 RESEARCH FRAMEWORK

The framework is a single causal pipeline, shared by the simulated and the real data: traffic is aggregated into fixed-length windows; each window is reduced to a feature vector; a detector consumes the feature stream, calibrating on an initial benign period and then scoring every subsequent window; and a thresholding stage turns scores into alarms, which are scored at window and event level. Using one pipeline for both data sources means the proposed detector and all baselines are exercised by identical code on simulated and on real traffic, so a difference in results reflects the data, not the plumbing. Figure 1 illustrates the idea: benign traffic occupies a band of normal variation that drifts slowly over time, and anomalies are the points that breach the band. A good detector tracks the band as it drifts and flags the breaches without alarming on ordinary variation inside it.

4.2 DATASET

Simulated data. Each trace is 6000 one-second windows. Benign traffic is a slow diurnal cycle multiplied by a first-order autoregressive wander and per-window log-normal noise, with injected benign *flash crowds* — short multiplicative surges in packets and source counts

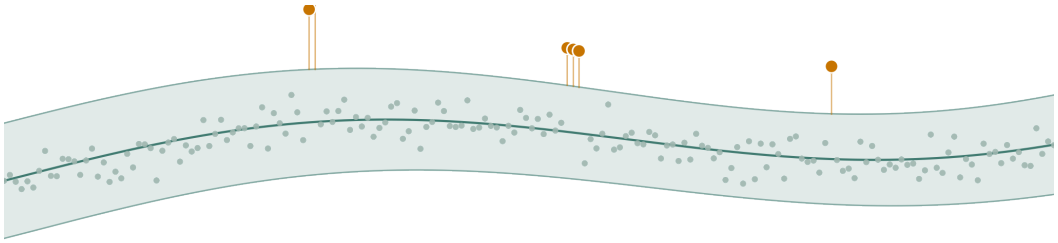


Figure 1.: The detection problem as a control chart. Benign traffic (grey) occupies a slowly drifting band of normal variation; anomalies (amber) are the points that breach it.

— so that the benign class contains realistic bursts, the hardest case for false-alarm behaviour. The first 600 windows are a guaranteed-benign calibration period. Fourteen anomaly events per trace are drawn, balanced across five families: a volume *burst*, a *SYN flood*, a *port scan*, a *protocol shift*, and a *slow ramp*. The first two are volumetric; the last three are non-volumetric and are the real test. An optional *drift* parameter adds a slow benign trend and random walk to the level.

Real data. We use CIC-IDS2017 [7], specifically the labelled flow CSVs (the archive `GeneratedLabelledFlows`, verified against its published MD5 checksum), which retain the timestamps, addresses, and ports needed to build time windows. We evaluate the Friday afternoon port-scan (150 windows, 10 attack) and DDoS (93 windows, 20 attack) captures.

4.3 DATA PREPROCESSING

For the simulator, preprocessing is the windowing itself. For CIC-IDS2017 each flow record is assigned to a time window by its timestamp. Because the Friday attack captures carry minute-resolution timestamps, the real-data window is one minute. Per window we sum forward and backward packets and bytes, count distinct destination ports and source addresses, sum SYN-flag counts, and sum the packets carried over UDP, producing exactly the six raw quantities the simulator produces. A window is labelled as attack when more than half of its flows are attack-labelled, which leaves ambiguous boundary windows benign. The two timestamp conventions in the dataset — second resolution on the benign Monday capture and minute-resolution, twelve-hour time on the Friday captures — are handled explicitly.

4.4 FEATURE SELECTION / EXTRACTION

From the six raw per-window quantities we derive an eight-dimensional feature vector: the logarithms of packets, bytes, distinct destination ports, and distinct source addresses; the SYN ratio and UDP fraction directly; and two ratio features, the logarithm of packets per source and the logarithm of bytes per packet. The logarithms stabilise the heavy-tailed counts. The ratio features are what expose non-volumetric attacks: a port scan inflates destination ports and packets per source while barely moving total volume, and a SYN flood collapses bytes per packet while raising the SYN ratio. The same eight features are computed identically for simulated and real traffic.

4.5 PROPOSED MODEL / ALGORITHM

The proposed detector maintains an online estimate of the mean $\mathbf{m}$ and covariance $\mathbf{S}$ of the feature vector and scores each window by its normalised Mahalanobis distance,

$$s_t = \sqrt{\tfrac{1}{p}\,(\mathbf{x}_t - \mathbf{m})^\top\, \tilde{\mathbf{S}}^{-1}\,(\mathbf{x}_t - \mathbf{m})}\,, \tag{1}$$

where $p = 8$ and $\tilde{\mathbf{S}}$ is $\mathbf{S}$ shrunk slightly toward its diagonal for numerical stability. After scoring, the estimates are updated by an exponentially weighted rule with rate α — but the update is *gated*. If s_t exceeds a gate, the window is treated as suspect: the mean is moved only by a small, winsorised step so a genuine sustained level shift is still eventually learned while a transient attack cannot drag the baseline toward itself, and the covariance update is skipped entirely, because the covariance update is quadratic in the deviation and even a single attack window would otherwise inflate the model's notion of normal spread. Below the gate, the ordinary EWMA update applies. The procedure runs in $O(p^2)$ time and keeps $O(p^2)$ state — 72 floating-point numbers and no stored history — and needs no training phase.

Five baselines bracket the proposed detector. Three are single-feature monitors on the log packet count, as a classic volume monitor would use: a *fixed threshold*, a *moving average* on the deviation from a sliding mean, and a *z-score* against a mean and standard deviation frozen at calibration. One is a strong multivariate baseline, a *static Mahalanobis* detector that freezes its mean and covariance at calibration. The last is the *Isolation Forest*, a heavier learning-based reference trained on the benign calibration period. Scores are turned into alarms by a single threshold; two optional post-filters (m-of-n persistence and a CUSUM accumulator) are evaluated in Section 9.

4.6 IMPLEMENTATION

The system is implemented in Python using NumPy and SciPy for the numerical work and scikit-learn for the Isolation Forest, in roughly seven hundred lines across the simulator, detectors, metrics, experiment harness, and real-data pipeline. A suite of fourteen unit tests covers simulator determinism, label correctness, the causality of every detector, the equivalence of the batch and streaming code paths, and the alarm post-filters. Figures are produced with Matplotlib. Reproducibility is enforced through a strict seed protocol: hyper-parameters and thresholds are tuned only on development seeds; a separate block of seeds used while debugging is never reported; and the headline numbers come from a final held-out block of seeds that the method never saw during development, run once after the method was frozen.

5. EXPERIMENTAL SETUP

All six detectors (Table 1) are evaluated by the shared pipeline. On simulated data each detector calibrates on the 600-window benign prefix and scores the rest; the alarm threshold for each is tuned on ten development seeds to maximise mean event F_1 subject to a benign false-positive rate below one percent. Results are reported on thirty fresh held-out seeds. We measure, per window, the detection (true-positive) rate and the false-positive rate; and, per event, the event recall, the event precision, their harmonic mean (F_1), and the detection delay. Confidence intervals are 95% t-intervals over the thirty seeds, and the proposed detector is compared with each baseline by a paired Wilcoxon signed-rank test over the same seeds, with Holm correction across detectors. Sweeps vary benign drift, attack intensity, benign noise,

Table 1.: The six detectors and their roles.

Detector	Features	Role
Fixed threshold	1 (volume)	baseline
Moving average	1 (volume)	baseline
Z-score (static)	1 (volume)	baseline
Mahalanobis (static)	8	strong baseline
Isolation Forest	8	heavy-ML reference
Adaptive (proposed)	8, online	proposed

Table 2.: Held-out mixed-anomaly results (mean over the final seeds, $\pm 95\%$ confidence interval). Best in bold.

Detector	Event F_1	FPR	False alarms / 1k
Fixed threshold	0.647 ± 0.026	0.002	1.00
Moving average	0.554 ± 0.022	0.003	1.05
Z-score (static)	0.639 ± 0.027	0.004	1.16
Mahalanobis (static)	0.917 ± 0.022	0.001	0.46
Isolation Forest	0.731 ± 0.027	0.007	3.80
Adaptive (proposed)	$\mathbf{0.924 \pm 0.016}$	$\mathbf{0.000}$	$\mathbf{0.11}$

and flash-crowd rate one at a time. On real data, where attack events are few, we additionally report the threshold-free area under the ROC and precision–recall curves. The method is frozen for the real data; only the alarm threshold is calibrated, on a benign period.

6. RESULTS

Table 2 and Figure 2 summarise the held-out mixed-anomaly scenario. The two multivariate detectors score around 0.92 event F_1, while every single-feature baseline sits between 0.55 and 0.65; the proposed detector also raises the fewest false alarms of any detector.

The per-attack-type breakdown (Figure 3) explains the gap. Volumetric events — floods and bursts — are caught by almost everything. Port scans and protocol shifts are caught only by the multivariate detectors and are completely missed by the volume monitors. Slow ramps are the hardest family for the proposed detector, because a gradual change is partly absorbed by its own adaptation; the Isolation Forest handles ramps best.

The decisive result is under non-stationarity (Figure 4). As benign traffic drifts, the static detectors degrade — static Mahalanobis falls from 0.917 to 0.796 and the volume monitors fall further — while the proposed detector stays essentially flat at about 0.92.

On real CIC-IDS2017 traffic (Figure 5, Table 3) the simulated finding reappears: on the non-volumetric port scan the multivariate detectors are near-perfect (ROC-AUC 0.999, recall 1.0) while the volume monitors miss most of it (recall 0.4). The DDoS capture is an easy volumetric case that even the fixed threshold detects perfectly.

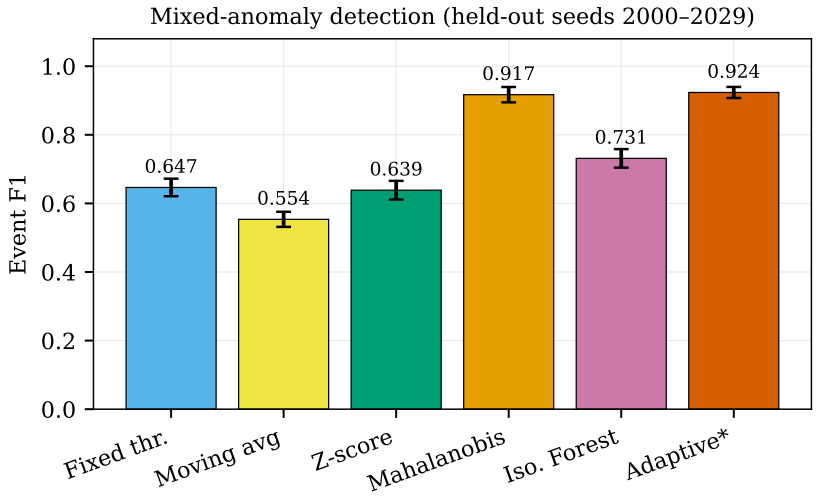


Figure 2.: Event F_1 by detector on the held-out mixed scenario. The multivariate detectors clear the single-feature baselines by a wide margin.

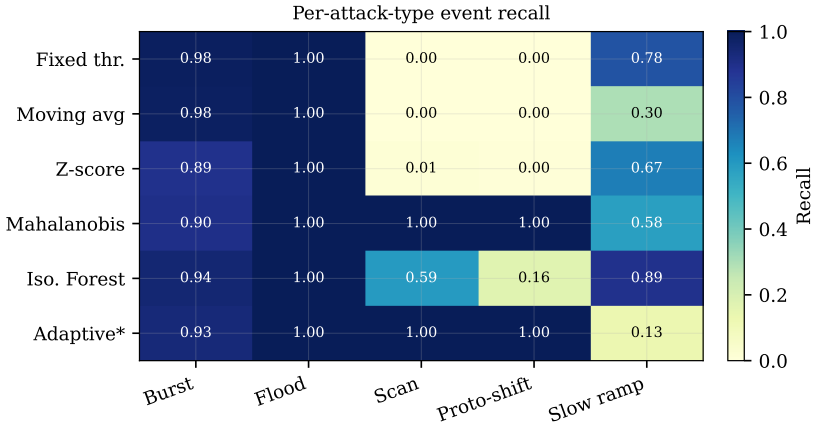


Figure 3.: Per-attack-type event recall. The multivariate detectors are the only ones that see port scans and protocol shifts; slow ramps are the proposed detector's weak point.

7. DISCUSSION

The results tell a consistent and deliberately narrow story. The large, reliable win is from using several features instead of one: both multivariate detectors see the attacks that volume monitors cannot, on simulated and on real traffic. This is a property of multi-feature statistical scoring in general, not of the proposed detector specifically, and we report it as such. The proposed detector's own contribution is narrower. In stationary traffic it is statistically tied with the much simpler static Mahalanobis baseline, so online adaptation is unnecessary when normal traffic does not move. Its value appears only when traffic is non-stationary, where a frozen model slowly mis-calibrates and drifts into false alarms while the online model tracks the benign change and holds its operating point — which is exactly the regime of a monitor calibrated once and left running for days. The slow-ramp weakness is the flip side of the same mechanism: adaptation that helps against benign drift also partly absorbs a slow attack.

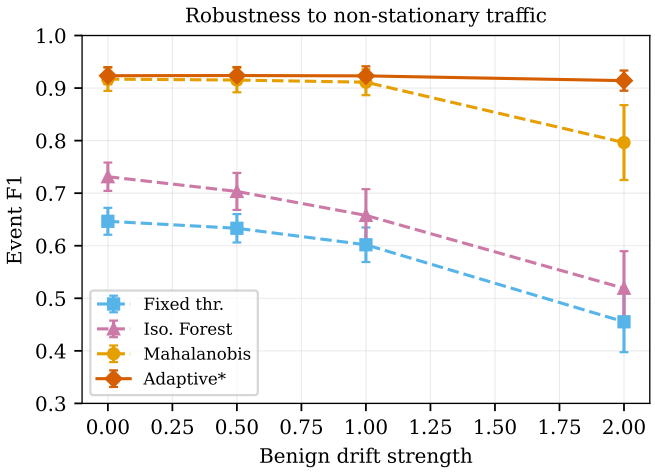


Figure 4.: Robustness to non-stationary benign traffic. The static detectors degrade as drift grows; the proposed online detector holds.

Table 3.:CIC-IDS2017 detection (method frozen; threshold calibrated on benign). Recall is measured at a 5% benign-calibrated operating point.

	PortScan		DDoS	
Detector	ROC-AUC	Recall	ROC-AUC	Recall
Fixed threshold	0.83	0.40	1.00	1.00
Moving average	0.68	0.40	1.00	1.00
Z-score (static)	0.80	0.40	0.87	0.00
Mahalanobis (static)	**0.999**	**1.00**	0.98	0.95
Isolation Forest	0.96	0.60	0.82	0.00
Adaptive (proposed)	**0.999**	**1.00**	0.98	0.95

8. COMPARATIVE ANALYSIS

Relative to the four single-feature baselines, the proposed detector is unambiguously better, by 0.19 to 0.37 event F_1 with significance at $p < 10^{-4}$ and a win on all thirty held-out seeds, because those baselines cannot represent non-volumetric attacks. Relative to the static Mahalanobis baseline it trades a statistically insignificant difference in stationary conditions (event F_1 0.924 versus 0.917, $p = 0.90$) for a clear, significant advantage under drift ($+0.118$, $p = 0.002$, winning on twenty-two of thirty seeds). Relative to the Isolation Forest — the only other multi-feature detector — it is both more accurate on the mixed scenario (0.924 versus 0.731) and far lighter in state.

Table 4 compares per-window cost and model state. The proposed detector keeps 72 floating-point values, about 167 times less than the Isolation Forest's ensemble, and needs no training. Its per-window compute of about $33\,\mu$s, dominated by an 8×8 matrix solve, is the highest of the six but still far inside a real-time budget for one-second windows — on the order of thirty thousand windows per second on a single core. "Lightweight" therefore holds on the dimensions that constrain a long-running monitor — memory and the absence of retraining — rather than on raw per-window arithmetic.

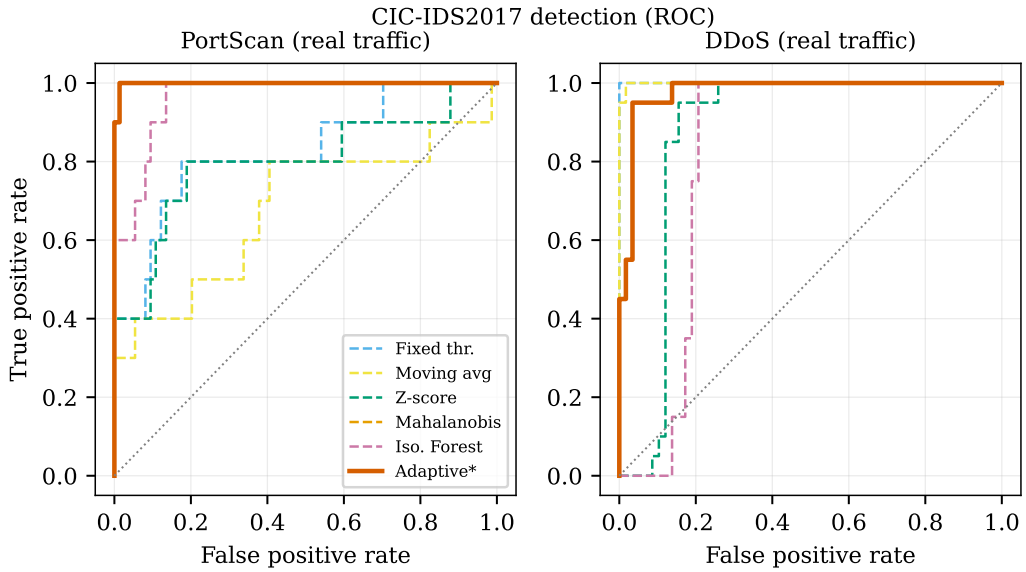


Figure 5.:ROC curves on real CIC-IDS2017 traffic. The multivariate detectors dominate on the port scan; the DDoS is easy for all volume-aware detectors.

Table 4.: Per-window compute and model state (lower is lighter).

Detector	Time (μs / window)	State (floats)
Fixed threshold	0.26	1
Moving average	0.66	60
Z-score (static)	0.30	2
Mahalanobis (static)	3.9	72
Isolation Forest	9.0	12,040
Adaptive (proposed)	32.8	72

9. ABLATION / STATISTICAL ANALYSIS

Figure 6 ablates the proposed detector's design on a stationary and a drifted scenario. Removing the gated update is the most damaging change (event F_1 falls from 0.91 to 0.82): without it, attack windows are absorbed into the model and the detector goes blind. Loosening the covariance gate so that more windows enter the covariance is almost as harmful, which confirms that freezing the covariance on suspect windows — not merely the mean — is necessary. Replacing the full covariance with a diagonal one costs most under drift ($0.916 \rightarrow 0.793$) and raises false alarms, showing that the correlations between features carry real signal. The two optional post-filters do not help: m-of-n persistence is marginally worse, and CUSUM is much worse ($\rightarrow 0.30$) because it accumulates on benign flash crowds and fires on them.

All comparisons in Sections 6 and 8 are supported by paired Wilcoxon signed-rank tests over the same thirty held-out seeds, with Holm correction applied across the five baselines within each setting. The proposed detector's advantage over the single-feature baselines is significant at $p < 10^{-4}$ in every setting; its difference from static Mahalanobis is not significant in the stationary case ($p = 0.90$) and is significant at the strongest drift ($p = 0.002$). The sensitivity sweeps over attack intensity and benign noise (not reproduced here for space) show the

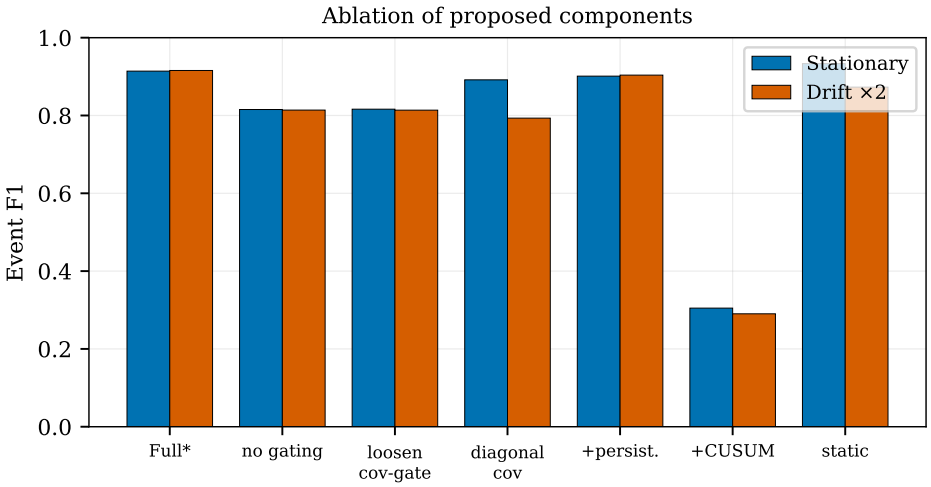


Figure 6.: Ablation of the proposed components in stationary and drifted traffic. The gated update and the full covariance carry the performance; persistence and CUSUM do not help.

multivariate pair on top throughout, with the proposed detector marginally behind static Mahalanobis at the highest intensities and marginally ahead under the heaviest noise.

10. LIMITATIONS

Several limits bound these claims. The controlled experiments use synthetic traffic, so absolute numbers reflect our generative assumptions rather than any particular production network; the real-data validation mitigates this but is itself limited. The CIC-IDS2017 captures are single, hour-long sessions with minute-resolution timestamps, which forces coarse one-minute windows, yields few attack windows, and is too short and too stationary to exhibit the drift on which the proposed detector's advantage rests — so that advantage is at present shown only in simulation. The detector also trails on very slow attacks, and its per-window matrix solve, while real-time here, would need an incremental-inverse implementation to scale to very high-dimensional feature sets.

11. CONCLUSION

We asked whether a lightweight, training-free statistical detector can catch the anomalies that volume monitors miss while staying robust to changing traffic. The answer is a qualified yes. Multi-feature statistical scoring is what defeats non-volumetric attacks such as port scans and protocol shifts, on both simulated and real traffic, and it does so with a few dozen parameters and no training. Online adaptation on top of that scoring is not needed in stationary conditions, where it ties the simplest multivariate baseline, but it is what keeps the detector accurate and calm when normal traffic drifts, which is the realistic regime for a monitor left running over days.

12. FUTURE WORK

The clearest next step is a real-data drift study on multi-day captures, which would let the drift advantage be measured outside simulation. The slow-ramp weakness invites a two-timescale variant that pairs a fast-adapting and a slow-adapting model so that gradual attacks are

not absorbed. Finally, an incremental update of the covariance inverse would remove the per-window matrix solve and let the feature set grow without raising cost.

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CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

DATA AVAILABILITY

The CIC-IDS2017 dataset is publicly available from the Canadian Institute for Cybersecurity. All simulation code, the real-data processing pipeline, result tables, and figure-generation scripts produced for this project are available in the accompanying project repository.

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